

Unit 10 Test Study Guide

(Rational Expressions & Equations)

Name: _____

Date: _____ Per: _____

Topic 1: Simplifying Rational Expressions

Directions: Simplify each expression.

1. $\frac{16xy^5}{42x^2y^3}$

2. $\frac{r^2 + 2r}{5r + 10}$

3. $\frac{6k^2 - 11k + 4}{9k^2 - 16}$

4. $\frac{y^3 - y}{y^3 + 8y^2 - 9y}$

Topic 2: Operations with Rational Expressions

Directions: Perform the indicated operation. Give each answer in simplest form.

5. $\frac{5v^3}{7v^2} \cdot \frac{21v^2}{20v}$

6. $\frac{a^2 + a - 6}{a^2 - a - 2} \cdot \frac{a^2 + 5a + 4}{a^2 + 2a - 3}$

7. $\frac{10m + 10}{8m^2 + 12m} \cdot \frac{2m^2 + m - 3}{m^2 - 1}$

8. $\frac{6q^5}{8q^5} \div \frac{9q}{8q^2}$

9. $\frac{c^2 - 6c + 8}{c^2 - 2c} \div (3c - 12)$

10. $\frac{p^2 - 36}{2p^2 + 3p + 1} \div \frac{4p - 24}{8p + 4}$

$$11. \frac{3x+10}{x^2-49} + \frac{x+18}{x^2-49}$$

$$12. \frac{n^2+7n}{n^2+4n-32} - \frac{n+16}{n^2+4n-32}$$

$$13. \frac{5}{18z} + \frac{1}{2z}$$

$$14. \frac{6r-2}{10r-20} - \frac{1}{r-2}$$

$$15. \frac{k^2-10k+1}{5k^2-2k-3} + \frac{1}{k-1}$$

$$16. \frac{v+1}{v+5} + \frac{5v-11}{v^2+4v-5} \div \frac{v^2-v-12}{v^2+2v-3}$$

Topic 3: Rational Equations

Directions: Solve each equation.

$$17. \frac{x+2}{6} = \frac{2x-10}{8}$$

$$18. \frac{r+1}{4} = \frac{14}{r}$$

$$19. \frac{8}{3a} = \frac{a}{6}$$

$$20. \frac{w-7}{6} = \frac{4}{w+3}$$

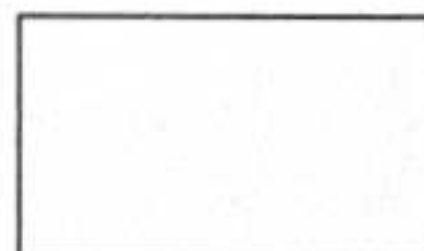
$$21. \frac{2}{c-3} - \frac{1}{4c-12} = \frac{2}{c-5}$$

$$22. \frac{7}{12y} - \frac{1}{6y} = \frac{y-1}{y+1}$$

Topic 4: Applications with Rational Expressions

23. Write an expression to represent the **area** of the rectangle.

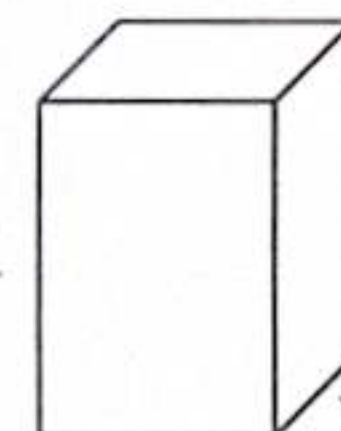
$$\frac{x^2 - 8x + 16}{4x^2}$$



$$\frac{2x^2 - 10x}{x^2 - 9x + 20}$$

24. Write an expression to represent the **volume** of the rectangular prism.

$$\frac{3r^2 + 3r - 60}{5r^2 + r}$$

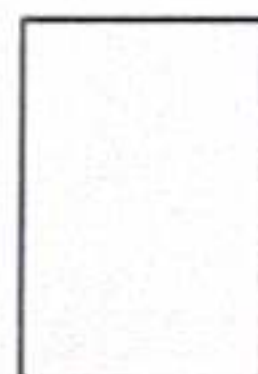


$$\frac{r^2 + r - 12}{r + 5}$$

$$\frac{10r^2 + 2r}{r^2 - 16}$$

25. Write an expression to represent the **perimeter** of the rectangle.

$$\frac{k^2 - 5}{k + 4}$$



$$\frac{3k + 1}{k + 4}$$

26. Write an expression to represent the **area** of the trapezoid.

